

Tsunami source of the 2011 Tohoku-Oki earthquake, Japan: Inversion analysis based on dispersive tsunami simulations

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[1] The large tsunami of the 2011 Tohoku-Oki earthquake was clearly recorded by the ocean bottom pressure and GPS wave gauges deployed in and around Japan. We estimated the initial tsunami water height distribution by inversion analysis of the waveforms based on dispersive tsunami simulations. The distribution is characterized by a peak height of 8 m located near the trench and the high-water (>2m) region extending landward with a width of ~100 km. A series of numerical simulations suggests that a relatively steep peak located near the trench is necessary in order to simultaneously reproduce the dispersive wave at a far-field station and the near-field waveforms. Furthermore, we estimated the coseismic slip distribution at the plate boundary, which indicates that large slip (~30 m) occurred at a depth of 20 km, which corresponds to a large slip deficit area in the interseismic period. Another slip (~25 m) occurred at the shallower part (<10 km) during the rupture.

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1. Introduction

[2] On March 11, 2011, the 2011 Tohoku-Oki earthquake occurred off the Pacific coast of northeastern Honshu, Japan. The event was a dip-slip rupture of the plate boundary between the Pacific and North America plates, and the moment magnitude (M_W) of this event was 9.0, which is the largest value ever recorded in Japan. This gigantic event excited a huge tsunami, which struck the Pacific coast of Japan resulting in nearly 20,000 people dead or missing. The tsunami field survey by the 2011 Tohoku Earthquake tsunami joint survey group [Mori *et al.*, 2011] indicates that the tsunami was significantly larger than that generated by the 1896 Sanriku tsunami earthquake (M_W 8) [Tanioka and Satake, 1996], and was comparable to the tsunami generated by the 896 Jogan earthquake, which occurred 1,100 years ago [e.g., Satake *et al.*, 2008].

[3] Analyses have indicated that large slip near the Japan Trench caused serious tsunami disasters as the result of the 2011 Tohoku-Oki earthquake [e.g., Fujii *et al.*, 2011; Maeda *et al.*, 2011]. Based on the seismogram analysis, Ide *et al.* [2011] insisted that dynamic overshoot, or slip larger than predicted by a static model, occurred at a very shallow

location during the rupture. Most previous tsunami waveform analyses used linear long-wave equations that cannot correctly simulate short-wavelength tsunami. An exception is a study by Lay *et al.* [2011] who conducted dispersive tsunami simulations. Dispersive simulations are more suitable to reliable source estimation than linear long-wave equations.

[4] Ocean-bottom pressure gauges deployed offshore are useful in tsunami source studies because they are free from the strong site effects in a very shallow bathymetry [Ritsema *et al.*, 1995] and thus can clearly observe tsunami signals including short-wavelength dispersive tsunami [González and Kurikov, 1993; Saito *et al.*, 2010b]. The ocean-bottom pressure gauges are usually located outside the focal area. However, in the case of the 2011 Tohoku-Oki earthquake, the ocean-bottom pressure gauges deployed inside the focal area were the first gauges to clearly observe the pressure change associated with tsunami from a giant earthquake [Ito *et al.*, 2011]. A significant improvement in resolution of the tsunami source is expected by analyzing these records obtained inside the focal area.

[5] The present study analyzes tsunami waveforms based on dispersive tsunami simulations in order to estimate the source of the disastrous tsunami associated with the 2011 Tohoku-Oki earthquake. We indicate the importance of the near-field tsunami records and dispersive waves at the far field for reliable and high-resolution source estimation. The estimated slip distribution on the plate boundary indicates that large slip occurred at a depth of 20 km, which corresponds to a large slip-deficit area in the interseismic period, and another large slip occurred at a very shallow location near the trench.

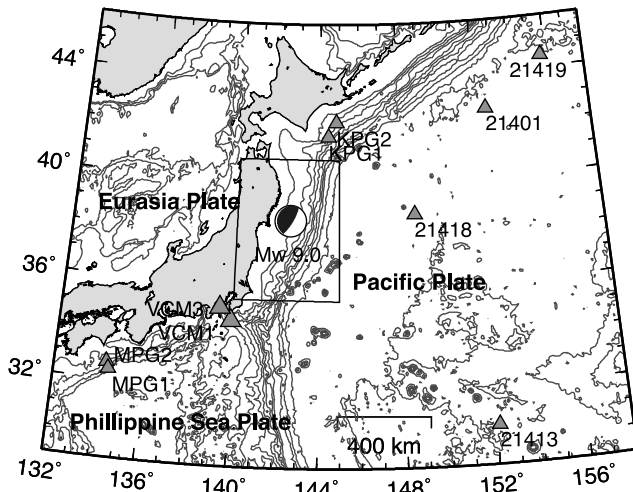
2. Tsunami Data

[6] The locations of ocean-bottom pressure and GPS wave gauges used in this study are indicated by triangles in Figure 1. The National Oceanic and Atmospheric Administration (NOAA) operates stations 21419, 21401, 21418, and 21413 as the DART system [Titov *et al.*, 2005]. The Japan Agency for Marine Earth Science and Technology (JAMSTEC) operates stations KPG1, KPG2, MPG1, and MPG2. The National Research Institute for Earth Science and Disaster Prevention (NIED) operates stations VCM1 and VCM3. In addition to these stations, we used the records obtained within the aftershock area of the 2011 Tohoku-Oki earthquake. Stations 801, 802, 803, 804, and 806 are the GPS wave gauges deployed by the Port and Airport Research Institute (PARI). We also used the pressure gauges of P02 and P06, which are temporal stations that were installed and operated between June 2010 and May 2011 by Tohoku University. The ocean-bottom pressure gauges measure the pressure at the ocean bottom, which is directly related to sea depth

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(a)



(b)

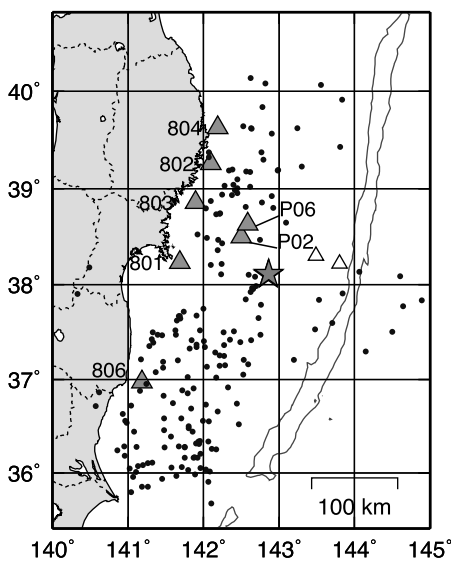


Figure 1. (a) Map of Japan and its surroundings. CMT solution of the 2011 Tohoku-Oki earthquake reported by USGS and the locations of the stations (gray triangles) used in this study. (b) Map of the focal area and its surroundings. The epicenter of the main shock (star) and aftershocks (dots) occurring within 24 hours of the main shock as determined by JMA are plotted. The two points in which the sea-bottom vertical deformation was observed by *Ito et al.* [2011] are shown by open triangles.

variation. The GPS wave gauges measure the variation of the sea-surface from the surface at rest. The tsunami waveforms used in this study are shown in Figure 2.

3. Tsunami Waveform Inversion

[7] Taking the x -axis along the trench axis and the y -axis perpendicular to the trench axis, we assume the possible tsunami source area to be a square of 560 km in the x axis and 240 km in the y axis. We set the 130 ($=13 \times 10$) grid points within the area. The initial water height $\eta_i(x, y, t=0)$

for the i th grid point, is assumed to be given by the Gaussian function as a basis function, as follows:

$$\eta_i(x, y, 0) = m_i \exp \left[-\frac{(x - x_i)^2}{(L_x/2)^2} - \frac{(y - y_i)^2}{(L_y/2)^2} \right], \quad (1)$$

where the center of the i th subregion is located at (x_i, y_i) . We set L_x and L_y to 96 km and 48 km, respectively. For the tsunami propagation, we use 2-D linear dispersive tsunami equations [Saito *et al.*, 2010a],

$$\begin{aligned} \frac{\partial M}{\partial t} + gh \frac{\partial \eta}{\partial x} &= \frac{1}{3} h^2 \frac{\partial^2}{\partial x \partial t} \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) \\ \frac{\partial N}{\partial t} + gh \frac{\partial \eta}{\partial y} &= \frac{1}{3} h^2 \frac{\partial^2}{\partial y \partial t} \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) \\ \frac{\partial \eta}{\partial t} &= -\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \end{aligned} \quad (2)$$

The parameters M and N are the velocity components integrated along the vertical direction from the sea bottom to the sea surface, η is the water height from the sea surface at rest, h is the water depth, and g is the gravitational constant. The bathymetry data of ETOPO1 is used for the simulation [Amante and Eakins, 2009]. The simulated tsunami height represents the sea-surface height variation from the sea surface at rest, which is comparable to the tsunami records obtained by GPS wave gauges. On the other hand, in order to compare with the sea-depth variation recorded by ocean-bottom pressure gauges, we corrected simulated waveforms for the contribution of the permanent sea-bottom deformation. As a correction, we added a constant value to the simulated waveforms so that the waveforms start with the amplitude of zero. In other words, we assumed that the depth does not change when the sea-bottom deformation starts. The observed ocean-bottom pressure records support this assumption and the amplitude is zero when the elapsed time from the earthquake origin time is zero. In order to estimate the model parameter m_i in equation (1), we applied the inversion method to the tsunami records using the damped least-squares method.

4. Results

[8] Figure 3a shows the estimated initial water height distribution. The waveforms calculated from this source, which simulate the observed records well, are plotted in Figure 2. The estimated initial water-height distribution indicates that the peak with a height more than 8 m locates near the trench, and the high-water (>2 m) region extends to landward with a width of ~ 100 km. This height distribution is consistent with 5 m uplift at the two points in the frontal wedge estimated by *Ito et al.* [2011]; the model of the present study indicates 5 and 7.5 m for the two points.

[9] In order to examine the contributions of dispersive waves, we construct a smooth source model because the smoother source excites less dispersive waves. The same inversion analysis is conducted for a smooth source model represented by large basis functions ($L_x = L_y = 96$ km). Figure 3b shows the estimated height distribution, which is relatively smooth and simple compared to the distribution in Figure 3a. The waveforms calculated from this source can roughly simulate all of the records, but cannot simulate the

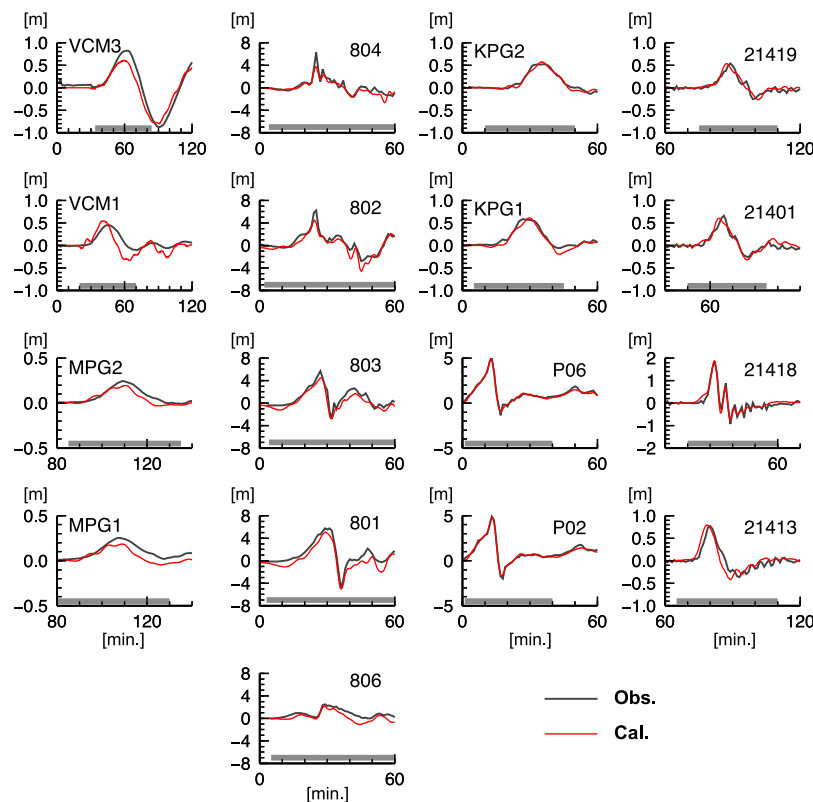


Figure 2. Tsunami waveforms for the 2011 Tohoku-Oki earthquake recorded by ocean-bottom pressure (KPG1, KPG2, VCM1, VCM3, P02, 21401, 21413, 21418, and 21419) and GPS wave gauges (801, 802, 803, 804, and 806). Observed and calculated waveforms are plotted by black and red lines, respectively. The horizontal axis indicates the elapsed time from the centroid moment time of the mainshock. The horizontal gray bars indicate the time range of data used in the inversion analysis. Dispersive tsunami equations are numerically solved based on the initial tsunami height distribution of Figure 3a.

record of station 21418, particularly, its later phases (Figure S1b in the auxiliary material), in which a dispersive tsunami is expected to arrive [Saito and Furumura, 2009].¹ Figures S1a and S1c show that dispersive tsunami equations successfully simulate the second peak of the waveforms, whereas the linear long-wave equations cannot. This confirms that the later phases in station 21418 are modeled by the dispersive waves.

[10] In order to investigate the contributions from the near field records of stations P02, P06, 801, 802, 803, 804, and 806, we conducted the same inversion analysis, but without using these seven near-field records. Figure 3c shows the initial water height distribution without using near-field records. This figure shows that the peak and crest are located landward. By definition, the initial water height distribution of Figure 3c simulates the far-field records well, but fails to simulate near-field records, in particular, the records of stations 804, 802, P06, and P02 (Figure S2).

[11] Summarizing the results of the above simulations, we conclude that a model of Figure 3a is necessary in order to fully explain the observed features of the dispersive wave at a far-field station and the near-field waveforms. Figure 3b is too smooth and simple to reproduce the dispersive waves. The comparison with Figure 3c suggests that near-field

records constrain the location of the peak and the crest near the trench.

5. Slip Distribution

[12] Various assumptions are necessary in estimating the slip distribution of the earthquake, such as the fault geometry and slip directions, although, at present, it is difficult to select the correct fault geometry from various models, and fault branching may occur in the rupture. We estimate the slip distribution by assuming that the slip occurs on the plate boundary with pure-dip slip. The 1-D plate boundary model constructed based on the 2-D model [Nakajima and Hasegawa, 2006] is used. We set the slip distribution as the 130 basis functions given by equation (1), calculated the sea-bottom uplift assuming homogeneous elastic half space and the initial tsunami height distribution from the sea-bottom uplift assuming instantaneous rupture in the constant water depth of 6 km based on the water-wave theory [Kajiura, 1963]. Figure 3d shows the slip distribution estimated from the waveform inversion analysis (impulse response tests are shown in Figure S3). The seismic potency of this distribution is $9.5 \times 10^{11} \text{ m}^3$, and the moment magnitude (M_W) is 9.0, assuming a rigidity of 40 GPa. The main part of the slip (30-m slip) occurred at a depth of 20 km corresponds to the large slip-deficit area or the strongly-coupled area estimated from the GPS data analyses

¹Auxiliary materials are available in the HTML. doi:10.1029/2011GL049089.

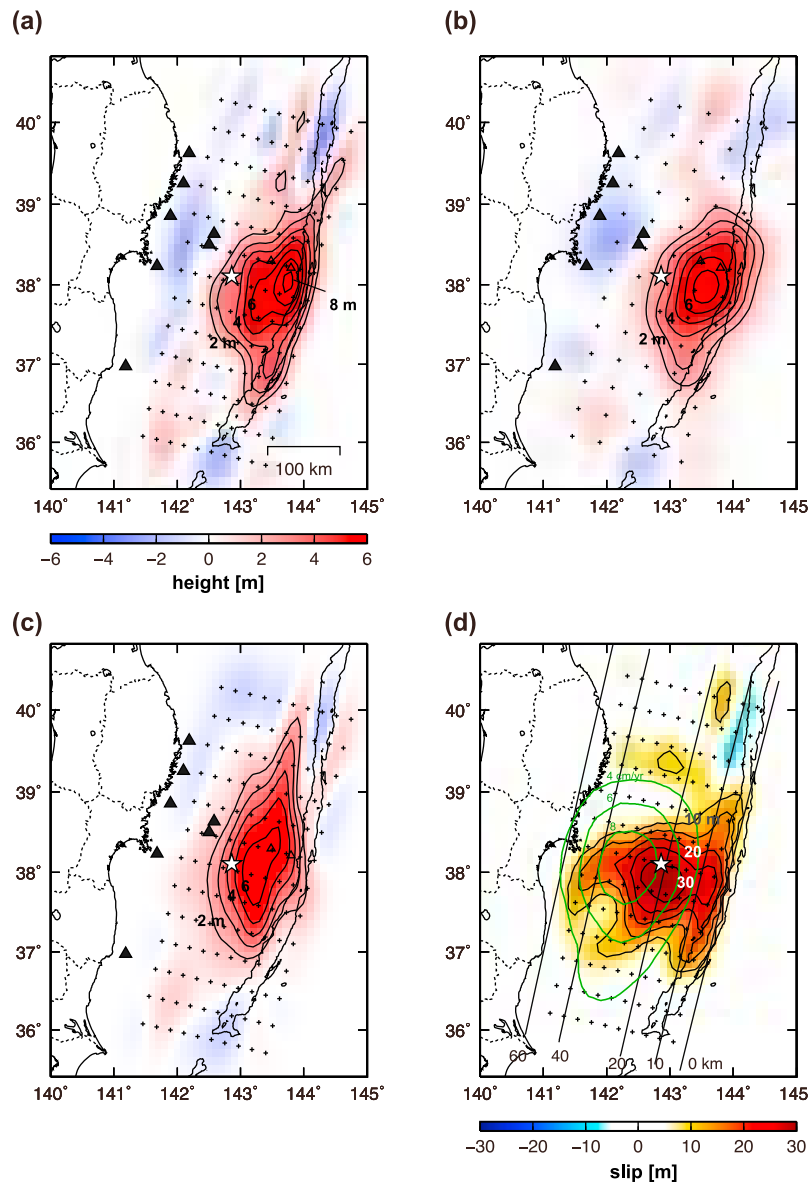


Figure 3. Initial tsunami height distributions and the slip distribution on the plate boundary estimated from the tsunami waveform inversion analyses. (a) The initial height distribution represented by 130 small basis functions estimated from all of the waveforms. The locations of the basis functions are plotted as crosses. The contours of the initial water height are plotted from 2 m to 8 m at intervals of 1 m. The epicenter of the main shock is plotted by a star. (b) Initial height distribution represented by 65 large basis functions estimated from all of the waveforms. (c) Initial height distribution represented by 130 small basis functions estimated from only the waveforms of VCM1, VCM3, MPG1, MPG2, KPG1, KPG2, 21401, 21413, 21418, and 21419. (d) Estimated slip distribution. The slip is assumed to occur on the plate boundary with pure dip slip. The contours of the plate boundary are plotted at depths of 0, 10, 20, 40 and 60 km. The contours of the slip distribution are plotted from 10 m to 30 m at intervals of 5 m. Strongly coupled area (back slip rate more than 4 cm/yr) estimated by *Nishimura et al.* [2004] is outlined by green contours.

before the Tohoku-Oki earthquake [Nishimura *et al.*, 2004; Suwa *et al.*, 2006; Hashimoto *et al.*, 2009]. On the other hand, the slip deficit corresponding to the shallower part (<10 km) of the slip (25 m) was not recognized. Dynamic fault rupture possibly causes larger slip near the surface than predicted by a static model [Mikumo and Miyatake, 1993; Aagaard *et al.*, 2001; Oglesby and Day, 2001; Ide *et al.*, 2011]. The relation between the strongly-coupled area before the earthquake and the coseismic distribution was also investigated for at the 2010 Chile earthquake [Moreno *et al.*,

2010; Lorito *et al.*, 2011; Vigny *et al.*, 2011]. In order to reveal the earthquake source process in greater details, not only tsunami data but also seismic and geodetic data must be analyzed assuming correct fault geometry.

6. Conclusion

[13] We estimated the initial tsunami water height distribution by the inversion analysis based on the dispersive tsunami simulations. The distribution is characterized by a

peak height of 8 m located near the trench and the high-water (>2m) region extending landward with a width of ~100 km. A series of numerical simulations suggests that a model of Figure 3a is necessary in order to fully explain the observed features of both the near-field waveforms and dispersive wave at a far-field station. In addition, the dispersive tsunami simulations are necessary in order to correctly reproduce short-wave length tsunami from the source. The estimated slip distribution at the plate boundary (Figure 3d) indicates that the large slip (~30 m) occurred at a depth of 20 km corresponds to a large slip-deficit area in the interseismic period. Another slip (~25 m) also occurred at a shallower location (<10 km) for tsunami excitation.

[14] **Acknowledgments.** We used records from offshore tsunami gauges operated by the Japan Agency for Marine-Earth Science and Technology (JAMSTEC), the National Oceanic and Atmospheric Administration (NOAA), the Port and Airport Research Institute (PARI), the National Research Institute for Earth Science and Disaster Prevention (NIED). Pressure measurements at P02 and P06 were operated under the MEXT project, "Evaluation and disaster prevention research for the coming Tokai, Tonankai and Nankai earthquakes". The study was also supported by JSPS KAKENHI (20244070). We used the earthquake catalog arranged by the Japan Meteorological Agency in cooperation with the Ministry of Education, Culture, Sports, Science and Technology. Careful reading of the manuscript and constructive comments by J. Pietrzak, R. Harris, and an anonymous reviewer were very valuable.

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